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2023026260101 45 | Jaya Vyas Lalit | DL - assignment - 1

Q1 Define an Artificial Neuron & list its components

A An Artificial Neuron is the basic processing unit of an Artificial Neural Network (ANN). It is inspired by biological neuron & receives one or more input signals, process them & produces an output.

Component of an Artificial Neuron:-

i Input $(x_1, x_2 \ldots, x_n)$: Values received from the previous layer

ii Weights $(w_1, w_2, \ldots, w_n)$: Determine the importance of each input

iii Bias (b): A constant value added to the weighted sum to improve flexibility

iv Weighted Sum: Sum of weighted inputs & bias

$$z = \sum_{i=1}^{n} w_i x_i + b$$

v Activation function: Produces the final output based on the weighted sum

Q2) Define weight & bias in an Artificial Neuron

A Weight: A weight is a numerical value assigned to each I/P of an artificial neuron. It determines the importance or influence of that input on the output

- Positive weight increases the influence
- Negative weight decreases the influence

Bias: A bias is an additional parameter added to the weighted sum before applying the activation function. It allows the neuron to shift the activation function. It allows the neuron to shift the activation function & improves learning

The mathematical equation is:

$$z = \sum_{i=1}^{n} w_i x_i + b$$

where w_i = weights, x_i = inputs, b = bias.

Q3 What is the role of activation function?

A) An activation function determines whether a neuron should be activated or not by converting the weighted sum into an output

Roles:

- Introduces non-linearity into the neural network
- Helps the network learn complex patterns
- Decides whether information should be passed to the next layer
- Enables deep neural networks to solve real-world problems

Q4 List two drawbacks of the Sigmoid Activation function.

145-B

A i) Vanishing Gradient Problem
for very large or very small i/p values, the gradient becomes very close to zero
- This slows down learning in deep neural networks

ii) Output is not zero-centered:
- The output ranges from 0 to 1
- This causes slower convergence during training because gradients are always positive.

Qs Define Vanishing Gradient Problem

A Vanishing Gradient Problem occurs during the training of deep neural networks when the gradient become extremely small as they are propagated backward through the layers.
As a result:
- Early layers learn very slowly
- Training becomes inefficient
- The model may fail to learn complex patterns

→ This Problem commonly occurs when using activation function like Sigmoid Eg Tanh

→ It can be reduced by using:
ReLU activation,
Batch Normalization,
ResNet.

Q6 Define Auto Encoder

145-B

A An autoencoder is an unsupervised neural network used to learn efficient representation of input data by compressing it into a lower-dimensional form & then reconstructing it

It consists of 2 main parts

i Encoder - Compresses the input into a latent representation

ii Decoder - Reconstructs the original input from the compressed representation

Auto Encoders are mainly used for:
Image denoising, Data compression, feature extraction, anomaly detection

Q7 Define a Convolutional Neural Network (CNN)

A A convolutional Neural Network (CNN) is a type of deep learning model specially designed for processing images & other grid-like data.

CNN automatically learns important features from image & other grid-like data

CNN automatically learns important features from images using convolution operations

Main Components:
Convolution layer,
Pooling layer, fully Connected layer.

Application of CNN include.

face recognition, medical image analysis, Object detection, self driving cars.

Qs Define feature Map in CNN

A feature Map is the output produced after applying a convolution filter (Kernel) to the input image in a CNN.
It highlights important features such as edges, textures shapes & patterns detected by the filter.

Each filter generates one feature map & multiple filters produce multiple feature maps.

Eg If a CNN uses 32 filters, it produces 32 feature Maps.

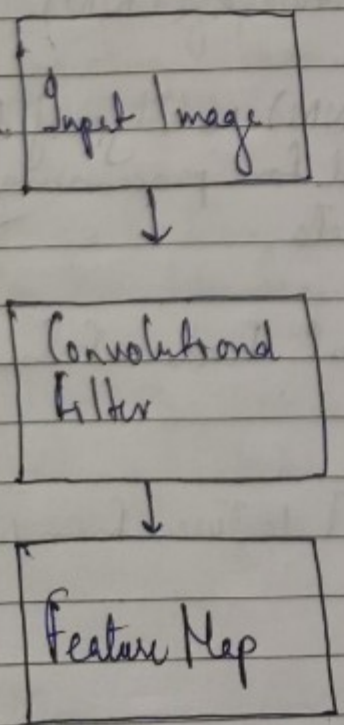


fig: Feature map (how to achieve)

Q9 What is Pooling?

A Pooling is an operation in a CNN that reduces the size of the feature map while preserving the most important information

Purpose:
- Reduces computation
- Improves feature extraction
- Decreases memory usage
- Prevents overfitting

Types of pooling

i Max Pooling - Selects the maximum value
ii Average Pooling - Computes the average value
iii Min Pooling - Selects the minimum value

Pooling makes CNN faster & more efficient

Q10 Explain the role of flattening layer.

A The flattening layer converts the 2-dimensional feature maps produced by convolution & pooling layer into a 1-dimensional vector.
The vector is then passed to the fully connected (dense) layer for classification

Role of flattening:
- Converts 2D flattening feature maps into 1D data
- Connects convolution & layers with dense layers.
- Prepares features for final classification.

- Enables the neural network to make prediction

eg A feature map of size : 4x4x3
↳ flattened into 48 x 1 vector
which is then given to the fully connected layer

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Deep Learning.

Assignment-1